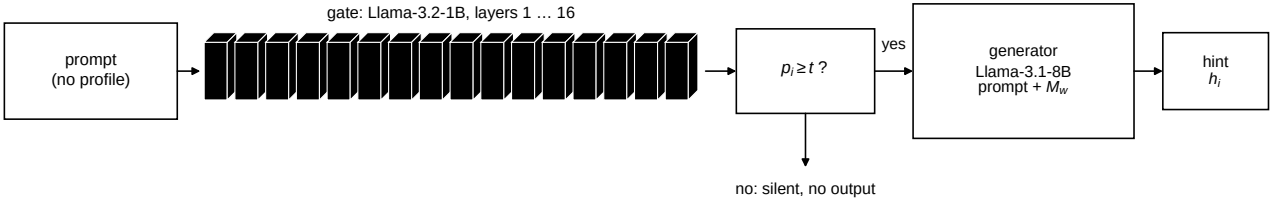
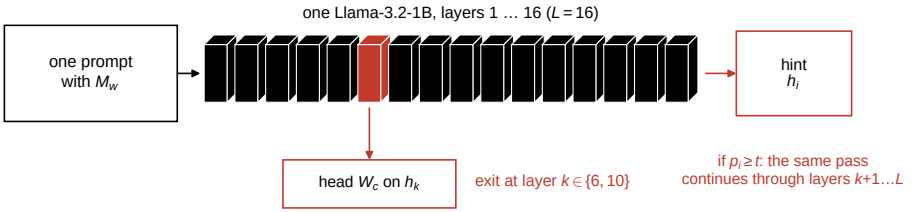
(a) Two-model cascade (LlamaPIE-style, our reimplementation)



silent marker: one full 1B pass. triggered marker: one 1B pass plus one full 8B pass. two base models resident.

(b) Unified early-exit decoder (this work)



silent marker: only layers 1 … k. triggered marker: exactly one pass. one base + one shared adapter, up to 1.05 GB resident.